Submission Summary

Conference Name
IEEE Chandigarh Subsection International Conference

Track Name
Artificial Intelligence and Data Science

Paper ID
1131

Paper Title
Dynamic AI Inference Offloading for Latency and Energy Optimization

Abstract
Contemporary AI-driven applications frequently depend on cloud-based inference, which can introduce delays, raise privacy concerns, and rely heavily on stable network connectivity. Although executing models directly on devices can mitigate latency and enhance data privacy, such approaches are limited by the restricted computational capabilities of mobile hardware. This paper proposes a hybrid edge–cloud inference framework that adaptively determines the most suitable execution location based on real-time system conditions and user-defined privacy requirements.
The framework incorporates a decision mechanism that accounts for both resource availability and privacy considerations by continuously tracking device metrics such as processor load, memory usage, thermal status, and network performance. Based on these inputs, the system decides whether to process inference locally or delegate it to a remote server. A unified execution layer supports smooth transitions between on-device models and cloud-based services, enabling flexible and efficient operation.
Experimental results indicate that the proposed approach achieves lower response times and more stable performance compared to static execution methods, while preserving user privacy preferences. Overall, the framework offers a practical and scalable solution for deploying intelligent applications on mobile devices with limited resources.

Created
2026-04-29 11:10:40 GMT+5:30

Last Modified
2026-04-29 11:10:40 GMT+5:30

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Submission Files

2210991889,2210992492_ResearchPaper.pdf (507.1 Kb, 2026-04-29 11:09:20 GMT+5:30)

Submission Questions Response

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3. Author's Information

I confirm that I have listed the title of the paper and the names, email addresses and affiliations of all contributed authors in the submission system that match the order as given in the submitted paper

Agreement accepted

4. Co-author's Consent

I confirm that I have discussed about this submission with all co-authors and authors have approved the paper and agree with its submission to this conference.

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5. Certificate of Originality

I confirm that neither the paper nor any parts of its content are currently under consideration or published in another conference/journal and the paper is free from plagiarism

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6. Author's Consent

It is confirmed that at least one author will register in full to attend and present the paper at the Conference.

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7. The use of artificial intelligence (AI)-generated text (e.g., by ChatGPT)

I confirm that no text in the paper is AI-generated.

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